U-FilSim: Pseudo Simulator by Video Diffusion Model with Uncertainty Filtering for Stable Reinforcement Learning

Anonymous Authors

***Abstract*—A video diffusion model is used as a pseudo simulator for reinforcement learning (RL). The main challenges are deviations from physical laws (hallucinations) and non-reproducibility due to stochastic generation, both of which can destabilize RL training. We propose U-FilSim, which defines uncertainty as latent-space ensemble variance and uses it as a confidence indicator reflecting physical correctness and dynamic predictability. Based on this uncertainty, high-uncertainty actions are filtered during training and execution, and a representative prediction is selected via a latent-space medoid. Experiments on PHYRE show that uncertainty correlates with physical deviation and dynamic unpredictability, and RL results measured by AUCCESS demonstrate the effectiveness of U-FilSim, particularly on more difficult tasks.**

***Index Terms*—video diffusion model, world model, uncertainty, pseudo simulator, reinforcement learning**

I. Introduction

Deep reinforcement learning (RL) has become a central methodology across diverse application domains, including robot control [1] and autonomous driving [2]. In the RL framework, an agent interacts with an environment and learns an action policy through trial and error. This learning process relies on a feedback loop: the agent selects an action based on an observation of the current state; the environment returns a reward; and the state transitions to a new one.

In real-world systems, however, trial-and-error learning can be costly in terms of time, wear, and safety, especially for robotic and other physical control tasks. To mitigate these risks, physics-engine simulators such as MuJoCo [3] are widely used during training, and learned policies are subsequently transferred to real environments. Building high-fidelity simulators, however, often requires substantial effort for modeling and parameter tuning, including object geometry, mass, friction coefficients, and other material properties [4]. Moreover, it remains difficult to accurately simulate unknown objects or operation in previously unseen environments.

Recently, large-scale video generation models—exemplified by Sora [5] and Wan 2.2 [6]—have been explored as world models or pseudo simulators. In this setting, the current visual observation and a candidate action are provided as input, and a video diffusion model generates a future observation sequence. Rewards and state transitions can then be estimated from the generated future sequence. A key advantage of this approach is that it can capture diverse dynamics directly from visual data, avoiding explicit and detailed physical modeling.

A fundamental concern, however, is whether diffusion-based video generation provides sufficient physical correctness and consistency to serve as a training environment for real-world RL. Because diffusion models learn data distributions rather than enforcing physical laws, they may generate physically implausible temporal evolutions or hallucinations [7]. In addition, their stochastic nature can produce different futures for the same input, introducing non-deterministic transitions that may destabilize RL training. This issue becomes more pronounced in environments with complex interactions, where a single initial state can lead to a wide range of predicted trajectories, reducing reliability as a simulator.

This paper develops a reliable pseudo simulator for RL training based on a video diffusion model, with explicit quantification of uncertainty arising from stochastic generation. Our key hypothesis is that uncertainty in the model's predicted dynamics is related to the stability of physically plausible trajectories. We characterize the uncertainty (i.e., unreliability) of model outputs by the diversity of generated videos conditioned on the same state–action input. Importantly, our experimental evaluation shows that this uncertainty quantity functions not merely as a measure of sampling fluctuation, but as a confidence indicator that reflects both (i) physical correctness, including the presence or absence of hallucination and violations of physical laws, and (ii) dynamic unpredictability, such as branching outcomes induced by collisions. Based on this interpretation, we filter out high-uncertainty actions and use only high-confidence actions for learning and execution, thereby improving the reliability of video diffusion models as pseudo simulators.

To address these issues, we propose Uncertainty-Filtered Simulator (U-FilSim), which consists of:

1) Uncertainty estimation for a candidate action in a given state using latent-space diversity across an ensemble of generated videos;
2) Training stabilization via uncertainty-based action filtering;
3) Consistent dynamics selection via medoid selection in latent space.

Overall, U-FilSim enables the use of a video diffusion model as a pseudo simulator while providing a more stable and reliable training environment for RL agents.

II. Related Works

A. World Models and Neural Simulators

World Models [8] were proposed to improve the efficiency of planning and policy learning by learning a compact representation of environment dynamics. In this framework, an agent encodes observations into a latent representation and predicts future observations through latent-state transitions, enabling imagined future prediction with reduced interaction in the real environment. Dreamer [9] advances this direction with a recurrent state-space model and achieves strong performance across visual control tasks, but long-horizon prediction quality can still degrade when the learned dynamics are inaccurate or overly compressed.

More recently, large-capacity generative models have enabled neural simulators that directly generate action-conditioned future observations. For example, UniSim [10] generates videos conditioned on actions, and large-scale video models such as Sora [5] show high visual realism with seemingly plausible motion. However, RL training requires more than realism: the pseudo simulator must provide physically reliable and stable feedback under repeated queries. Ensuring such reliability—especially in the presence of stochastic generation—remains an open problem. This work addresses that gap by improving the reliability of diffusion-generated future predictions when they are used as a pseudo simulator for RL.

B. Uncertainty Estimation for Diffusion Models

Diffusion models such as DDPMs [11] generate samples by reversing a gradual noising process. Because generation is inherently stochastic, identical conditioning inputs can yield different outputs. Prior studies have explored uncertainty in diffusion generation—for example, via Bayesian formulations or sample filtering—to detect unreliable outputs and improve generation quality [12], [13]. In these approaches, uncertainty is mainly used as a proxy for fidelity to the conditioning signal or overall sample quality.

In contrast, our goal is RL-oriented: we use uncertainty as a confidence indicator for predicted dynamics. We define uncertainty by the diversity of generated videos under the same state–action input, and interpret high uncertainty as reflecting either physical incorrectness (hallucinations/violations of physical laws) or dynamic unpredictability (e.g., branching outcomes due to collisions). Based on this uncertainty, we filter low-confidence actions during training and execution, and select a representative prediction (medoid) to provide consistent simulator feedback.

C. PHYRE

PHYRE [14] is a benchmark for physical reasoning based on a deterministic 2D physics engine. It provides physics puzzles governed by classical mechanics and supplies physically correct trajectories for a given action, which can be treated as ground truth. In this work, PHYRE is used to evaluate the physical reliability of diffusion-based pseudo simulation and the effectiveness of uncertainty-based filtering in RL.

III. Proposed Method

This section describes the components and algorithms of U-FilSim (Uncertainty-Filtered Simulator). Fig. 1 shows the overview. The objective is to improve the reliability of a video diffusion model when it is used as a pseudo simulator for RL. As motivated in the introduction, diffusion-based video prediction may produce (i) physically incorrect dynamics (hallucinations or violations of physical laws) and (ii) dynamically unpredictable outcomes (e.g., branching due to collisions). U-FilSim addresses both issues by estimating uncertainty from an ensemble of generated videos and using this uncertainty as a confidence indicator to filter unreliable actions. For actions that pass the filter, U-FilSim returns a single representative future sequence to provide consistent environment feedback to the RL agent.

Three main components are (1) Future video prediction using a video diffusion model, (2) Uncertainty (confidence) estimation using ensemble diversity in latent space, and (3) Uncertainty filtering and representative prediction selection via a latent-space medoid [15].

The following subsections detail each component.

A. Future Video Prediction with a Video Diffusion Model

We adopt the image-to-video diffusion model Wan 2.2 IT2V-5B [6] as the backbone of the pseudo simulator. Wan 2.2 is pretrained on large-scale general video datasets and then fine-tuned using video sequences generated by the PHYRE simulator.

The input and output of the model are given as follows:

Input
- $\boldsymbol{s}_0$: an image observation of a 2D environment.
- $\boldsymbol{a}$: an action selected by the agent. In our setting, $\boldsymbol{a}$ encodes the 2D position and size of an object to be placed into the scene. The corresponding object is inserted into $\boldsymbol{s}_0$ to form the conditional input.
- A fixed text prompt describing the physical setting (kept constant across samples).

Output $\boldsymbol{s}_{1:T} = f_\theta(\boldsymbol{s}_0, \boldsymbol{a})$: a predicted future image sequence of length T, where f_θ denotes the conditional video diffusion model.

We fine-tune the backbone model on PHYRE-generated videos using the conditional flow matching loss implemented in Wan 2.2. This training adapts f_θ so that, conditioned on $(\boldsymbol{s}_0, \boldsymbol{a})$, it reproduces the distribution of future sequences in the target environment.

B. Uncertainty Estimation as a Confidence Indicator

Although the diffusion model is conditioned on $(\boldsymbol{s}_0, \boldsymbol{a})$, its output is stochastic because generation involves sampling an initial noise variable. Therefore, for a fixed input $(\boldsymbol{s}_0, \boldsymbol{a})$, we generate an ensemble of N future video predictions by sampling N independent noise variables. We then compute uncertainty from the diversity of this ensemble. In our experiments, we set $N = 25$.

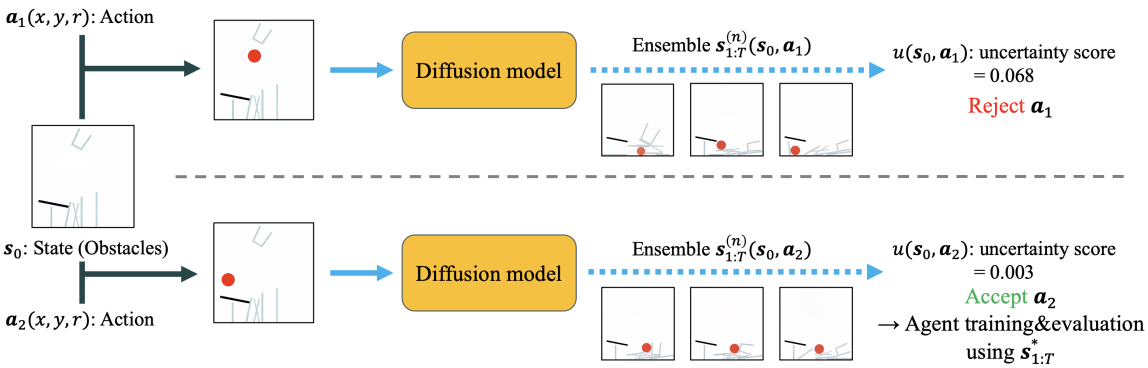

Fig. 1. Overview of U-FilSim. Given an initial observation $\boldsymbol{s}_0$ and a candidate action $\boldsymbol{a}$, a video diffusion model generates an ensemble of future video predictions by sampling different initial noises. Uncertainty is estimated from the ensemble diversity in latent space and used as a confidence indicator. **Top row**: a low-confidence (high-uncertainty) action. **Bottom row**: a high-confidence (low-uncertainty) action.

Let the n-th predicted future sequence be

$$\boldsymbol{s}^{(n)}_{1:T} = f_\theta(\boldsymbol{s}_0, \boldsymbol{a}; \epsilon^{(n)}), \tag{1}$$

where $\epsilon^{(n)}$ is the initial noise sample. Let $\boldsymbol{z}^{(n)}(\boldsymbol{s}_0, \boldsymbol{a}) \in \mathbb{R}^D$ denote the D-dimensional latent representation associated with $\boldsymbol{s}^{(n)}_{1:T}$.

We define the uncertainty $u(\boldsymbol{s}_0, \boldsymbol{a})$ as the channel-averaged variance of latent representations over the ensemble:

$$u(\boldsymbol{s}_0, \boldsymbol{a}) = \frac{1}{D} \sum_{i=1}^{D} \operatorname*{Var}_n\Big(z_i^{(n)}(\boldsymbol{s}_0, \boldsymbol{a})\Big) . \tag{2}$$

We compute diversity in latent space rather than in pixel space because latent representations are expected to capture dynamics-relevant factors more compactly, making diversity estimation less sensitive to superficial appearance changes.

Intuitively, A larger $u(\boldsymbol{s}_0, \boldsymbol{a})$ indicates lower confidence in the model's predicted dynamics for the given state–action pair. As shown in our experiments, this uncertainty reflects not only sampling fluctuation, but also (i) deviations from physical laws and (ii) inherent dynamic unpredictability. Accordingly, we use $u(\boldsymbol{s}_0, \boldsymbol{a})$ as a confidence indicator for action selection and simulator feedback.

C. Uncertainty Filtering and Representative Prediction Selection

U-FilSim uses the uncertainty $u(\boldsymbol{s}_0, \boldsymbol{a})$ in two steps:

1) Uncertainty Filtering: For each initial observation $\boldsymbol{s}_0$, we define the set of low-confidence (high-uncertainty) actions as

$$\mathcal{A}_{\text{high}}(\boldsymbol{s}_0) = \{\, \boldsymbol{a} \mid u(\boldsymbol{s}_0, \boldsymbol{a}) > \tau \,\}, \tag{3}$$

where τ is an uncertainty threshold.

During training: actions in $\mathcal{A}_{\text{high}}(\boldsymbol{s}_0)$ are labeled as failures so that the RL agent learns to avoid them (details are provided in the RL experimental setting).

During execution (test time): actions in $\mathcal{A}_{\text{high}}(\boldsymbol{s}_0)$ are removed from the candidate action set.

Rather than using a fixed global threshold, we set τ adaptively by selecting the top $p\%$ highest-uncertainty actions within each batch. This normalization mitigates scale differences in uncertainty across states and training stages.

2) Representative Prediction by Latent-space Medoid: Even for high-confidence actions, the diffusion model may generate slight variations across samples. To provide consistent environment feedback, we select one representative future sequence from the ensemble. We adopt a medoid criterion in latent space [15], which selects the sample whose latent is most central in the ensemble.

Let $d(\cdot, \cdot)$ be the Euclidean distance (L_2 norm) in the latent space. The medoid index is

$$i^* = \arg\min_i \sum_{j=1}^{N} d\Big(\boldsymbol{z}^{(i)}, \boldsymbol{z}^{(j)}\Big) , \tag{4}$$

and the selected future sequence is obtained by decoding the medoid latent:

$$\boldsymbol{s}^*_{1:\mathrm{T}} = \mathrm{Decoder}(\boldsymbol{z}^{i^*}) . \tag{5}$$

This procedure yields a single, representative prediction while remaining consistent with the ensemble distribution, thereby reducing non-determinism in the pseudo simulator.

IV. Experimental Setup for Reinforcement Learning

A. 2D Physics Environment

We use PHYRE [14] as a 2D environment governed by deterministic classical mechanics. For each episode, an initial observation $\boldsymbol{s}_0$ and an action $\boldsymbol{a}$ define the simulation, and PHYRE provides a ground-truth (GT) video of length $T = 49$ frames showing the resulting dynamics (e.g., falling under gravity with collisions). The scene contains fixed obstacles (black) and movable obstacles (gray), following the PhyWorld dataset [7].

Both GT videos (PHYRE) and predicted videos (video diffusion model) are represented as normalized RGB frame sequences of resolution 512×512.

1) Action Space: An action is a 3D continuous vector $\boldsymbol{a} = (x, y, r)$, indicating placement of a red disk with center (x, y) and radius r. The radius is limited to at most one quarter of the environment width, and the red disk must not overlap with obstacles. After placement, the red disk moves under gravity and interacts with obstacles via collisions.

2) Tasks: The agent selects $\boldsymbol{a}$ at the initial frame $\boldsymbol{s}_0$ to achieve the goal state by the final frame. We evaluate three difficulty levels:

Level 1 The red disk touches the floor (typically solvable via a direct fall).
Level 2 The red disk is inside the glass, requiring obstacle-aware placement.
Level 3 The red disk remains within a designated area, often requiring multi-object interactions (e.g., collision-induced motion).

B. Data and Model Training

1) RL Agent: We use the PHYRE DQN scoring model as the RL agent. Given $(\boldsymbol{s}_0, \boldsymbol{a})$, the agent predicts the success probability of the task. The model encodes $\boldsymbol{s}_0$ using a ResNet-18–based feature extractor and embeds $\boldsymbol{a}$, then outputs a scalar score.

For evaluation clarity, the agent does not directly optimize continuous actions. Instead, it scores actions from a predefined candidate list and attempts them in descending order of predicted success probability.

2) Reward: The reward is binary: success if the goal is achieved in the final frame; otherwise failure.

3) Action sets (train/val/test): For each task instance (fixed $\boldsymbol{s}_0$ and goal), we prepare candidate action sets of size 800/100/100 for train/validation/test, respectively. Each set contains at least one successful action. The validation and test action sets are strictly held out: they are never used for RL training and are not used to fine-tune the video diffusion model.

4) Fine-tuning the Pseudo Simulator: The video diffusion model (Wan 2.2 IT2V-5B) is fine-tuned using GT videos from the training action set. Conditioning follows the proposed setup: the model takes $(\boldsymbol{s}_0, \boldsymbol{a})$ (with the object inserted into $\boldsymbol{s}_0$) and a fixed text prompt describing the physical setting. Hyperparameters follow DiffSynth-Studio [16].

5) Training Data Generation for The RL Agent: To train the agent, each $(\boldsymbol{s}_0, \boldsymbol{a})$ in the training set is evaluated by either: the GT simulator (PHYRE), or the pseudo simulator (video diffusion model with U-FilSim).

From the resulting video prediction, success/failure is determined and used as the supervisory label for the DQN scoring model. The training objective is binary cross-entropy, and other settings follow the default PHYRE protocol.

C. Evaluation Metrics

We evaluate the experimental results from two perspectives:

1) Latent MSE for pseudo simulator fidelity: To quantify the difference between GT and pseudo-simulated predictions, we compare the resulting future video sequences produced from the same input $(\boldsymbol{s}_0, \boldsymbol{a})$. The discrepancy is measured as mean squared error (MSE) in the latent space of a VAE encoder, rather than in pixel space. This latent-space MSE is used as our fidelity metric for the pseudo simulator.

2) AUCCESS for RL agent performance: To evaluate the RL agent trained using simulation video datasets (from either GT or pseudo simulation), we use AUCCESS, the standard metric introduced in PHYRE. AUCCESS is defined as the area under the success-rate curve, where the success rate is plotted as a function of the number of attempted actions (trials). A higher AUCCESS indicates that the agent typically succeeds in fewer trials, i.e., it identifies effective actions more efficiently.

V. Experimental Evaluation of Uncertainty and Physical Correctness

Before the RL experiments in Sec. VI, this section reports two evaluations to validate whether the proposed uncertainty u functions as a confidence indicator for diffusion-based pseudo simulation. Specifically, we examine whether u reflects (i) physical correctness of the predicted dynamics, and (ii) dynamic unpredictability related to orbital stability (e.g., branching outcomes caused by collisions).

The positive result for (i) supports uncertainty-based filtering to remove physically incorrect predictions (hallucinations). In addition, the positive result for (ii) supports the use of u to detect hard-to-predict situations that may destabilize RL training, even when the predicted dynamics are physically plausible.

A. Evaluation I: Relation between Uncertainty and Physical Correctness

1) Objective: Evaluation I investigates whether uncertainty u reflects the degree to which a predicted video deviates from the physically correct dynamics, i.e., whether u can detect violations of physical laws (hallucinations).

2) Methods: First, the pseudo simulator (Wan 2.2) is fine-tuned using the training set described in Sec. IV. For each $(\boldsymbol{s}_0, \boldsymbol{a})$ in the test set, we generate a representative predicted video sequence $\boldsymbol{s}^*_{1:T}$ using U-FilSim (i.e., after ensemble generation and medoid selection). We then compute the uncertainty $u(\boldsymbol{s}_0, \boldsymbol{a})$ from the ensemble diversity in latent space.

Next, we obtain the ground-truth (GT) video sequence $s^{\mathrm{GT}}_{1:T}$ for the same $(\boldsymbol{s}_0, \boldsymbol{a})$ from the PHYRE simulator. To quantify physical deviation, we compute the L_2 distance (MSE) between $\boldsymbol{s}^*_{1:T}$ and $s^{\mathrm{GT}}_{1:T}$ in a VAE latent space. Importantly, the distance is evaluated over the entire time evolution, not only the final frame, to measure trajectory-level disagreement.

3) Results and Consideration: Fig. 2 plots uncertainty u against the latent-space L_2 distance from the GT trajectory. The correlation coefficient $r = 0.793$ indicates a strong positive correlation. Therefore, actions with high uncertainty tend to produce predictions that substantially deviate from GT dynamics, which corresponds to physically incorrect time

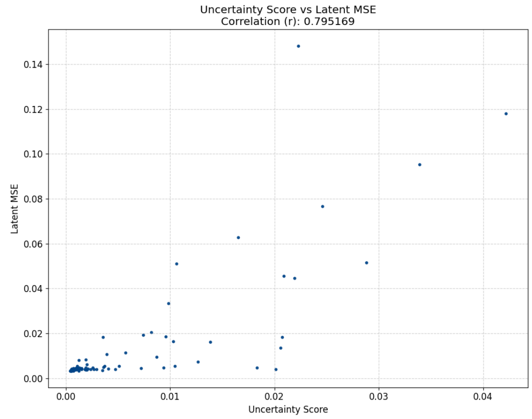

Fig. 2. Relationship between uncertainty u and the latent-space L_2 distance from the GT trajectory (PHYRE) on the test set.

evolution (hallucination). In contrast, actions with low uncertainty yield predictions that are close to GT, indicating higher physical correctness.

This result suggests that u is not merely a measure of sampling fluctuation: it functions as a confidence indicator that detects deviations from physical laws. Accordingly, U-FilSim filters out high-uncertainty state–action pairs so that physically unreliable predictions are not used for RL training.

B. Evaluation II: Relation between Uncertainty and Dynamic Unpredictability

1) Objective: Even when predicted dynamics are physically plausible, highly unpredictable situations can destabilize RL training and degrade task execution. Evaluation II examines whether uncertainty u captures such dynamic unpredictability, which is related to orbital stability and branching behavior (e.g., collision-induced divergence).

2) Experimental Settings: In the 2D environment described in Sec. IV-A, increasing the initial height y of the red disk generally increases the falling time and the number of possible collisions with obstacles. Because collisions can create branching outcomes, small differences in the trajectory can grow over time, leading to reduced predictability.

To test whether u reflects this effect, we evaluate uncertainty under a fixed environment while varying the initial height y of the red disk in the action $\boldsymbol{a} = (x, y, r)$. We group actions into three ranges of y and compare the resulting uncertainty distributions.

3) Results and Consideration: Fig. 3 shows the uncertainty distributions for three initial height ranges. As the initial height increases, the uncertainty values become larger and the distribution becomes wider. This indicates that u increases in situations that tend to produce collision-driven branching and reduced predictability.

Therefore, uncertainty u reflects not only physical correctness (as in Evaluation I) but also dynamic unpredictability associated with orbital instability. This supports the use of U-FilSim to exclude actions that are difficult to predict and may

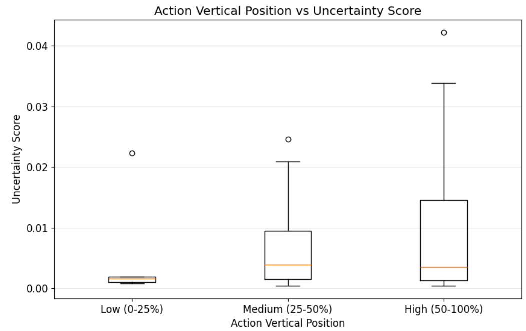

Fig. 3. Distributions of uncertainty u for three ranges of the red disk's initial height y in the action.

destabilize RL training, even if they are not strictly "incorrect" in a physics sense.

C. Summary

These two evaluations indicate that the proposed uncertainty u functions as a confidence indicator capturing both (i) physical correctness (deviation from GT dynamics) and (ii) dynamic unpredictability (branching/instability caused by collisions and long-term sensitivity). Supported by these results, U-FilSim filters out high-uncertainty state–action pairs to improve the reliability and stability of diffusion-based pseudo simulation for RL.

VI. Experiments of Reinforcement Learning

Table I summarizes RL performance measured by AUCCESS. Each DQN scoring model is trained using labeled outcomes obtained from either the GT simulator (PHYRE) or the pseudo simulator (Wan 2.2). For the pseudo simulator, we compare three types of training samples: single generation ($N = 1$), medoid selection from an ensemble ($N = 25$), and the proposed U-FilSim (medoid selection with top 20% high-uncertainty actions rejected). We also report the latent-space L_2 distance from GT, which quantifies physical deviation of the predicted dynamics.

Overall, the average AUCCESS across the three task levels is low with single generation ($N = 1$) and improves with medoid selection, indicating that reducing stochastic variation yields more reliable training signals. The proposed uncertainty filtering further improves performance, achieving an average AUCCESS of 0.924, which is close to the GT simulator. Notably, for the most difficult Level 3, U-FilSim exceeds the GT simulator. This suggests that uncertainty filtering is beneficial not only for removing physically incorrect predictions but also for detecting dynamically unpredictable situations that destabilize learning. By encouraging the agent to avoid such actions, U-FilSim can yield higher task success than training with the deterministic physics simulator.

The physical deviation measured by the latent-space L_2 distance from GT shows a similar trend. Both medoid selection and uncertainty filtering reduce the discrepancy from GT compared to single generation, indicating improved physical

TABLE I

COMPARISON OF AUCCESS AS PERFORMANCE OF RL AGENT TRAINED BY GT AND PSEUDO SIMULATORS. THREE TYPES OF TRAINING SAMPLES ARE $N = 1$, MEDOID WITH $N = 25$ AND UNCERTAINTY FILTERED. DEVIATIONS FROM GT ARE ALSO SHOWN. THE BEST PERFORMANCE IS SHOWN IN BOLD.

Agent	Simulator	Training samples	AUCCESS ↑				L_2 dis. from GT ↓
			Level 1	Level 2	Level 3	Avg.	
Random	-	-	0.285	0.101	0.059	0.148	-
DQN	PHYRE	-	**1.000**	**0.925**	0.877	**0.934**	0
DQN	Diffusion	Single sample	0.853	0.880	0.813	0.849	0.016
DQN	Diffusion	Medoid sample	0.916	0.910	0.872	0.899	0.011
DQN	Diffusion	Proposed sample	0.976	0.911	**0.885**	0.924	**0.005**

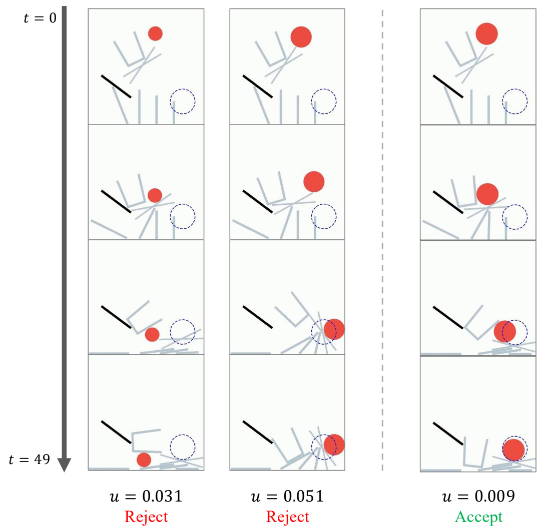

Fig. 4. Predicted future sequences are shown using four frames for Level 3 task. The left two actions are rejected (high uncertainty), and the right action is accepted (low uncertainty). The goal region is indicated by the blue circle at the bottom right.

consistency of the pseudo simulator outputs. This increased reliability aligns with the observed improvements in AUCCESS.

Fig. 4 provides qualitative examples of uncertainty-based filtering for Level 3 task. The two rejected actions (left) correspond to high uncertainty and produce unreliable or difficult-to-predict dynamics, whereas the accepted action (right) yields a more consistent and goal-directed trajectory. These examples illustrate how U-FilSim filters out low-confidence actions and concentrates training on state–action pairs with a higher probability of success.

VII. CONCLUSION

This paper studied the use of a video diffusion model as a pseudo simulator for reinforcement learning, focusing on two issues: hallucinations that violate physical laws and stochastic non-reproducibility that destabilizes training. We proposed U-FilSim, which estimates uncertainty as latent-space ensemble variance, uses it as a confidence indicator, selects a medoid prediction for consistency, and filters out high-uncertainty actions during training and execution.

Experiments in PHYRE showed that uncertainty correlates strongly with deviation from GT dynamics and also reflects dynamic unpredictability such as collision-induced branching. RL results demonstrated that U-FilSim achieves AUCCESS close to the physics simulator and can surpass it on difficult tasks by learning to avoid low-confidence actions.

Future work includes improving out-of-distribution robustness, reducing ensemble inference cost, and better calibrating the uncertainty measure.

REFERENCES

[1] C. Tang, B. Abbatematteo, J. Hu, R. Chandra, R. Martín-Martín, and P. Stone, "Deep reinforcement learning for robotics: A survey of real-world successes," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 8, no. 1, pp. 153–188, 2025.

[2] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. A. Al Sallab, S. Yogamani, and P. Pérez, "Deep reinforcement learning for autonomous driving: A survey," *IEEE transactions on intelligent transportation systems*, vol. 23, no. 6, pp. 4909–4926, 2021.

[3] E. Todorov, T. Erez, and Y. Tassa, "Mujoco: A physics engine for model-based control," in *2012 IEEE/RSJ international conference on intelligent robots and systems*. IEEE, 2012, pp. 5026–5033.

[4] W. Zhao, J. P. Queralta, and T. Westerlund, "Sim-to-real transfer in deep reinforcement learning for robotics: a survey," in *2020 IEEE symposium series on computational intelligence (SSCI)*. IEEE, 2020, pp. 737–744.

[5] T. Brooks, B. Peebles, C. Holmes, W. DePue, Y. Guo, L. Jing, D. Schnurr, J. Taylor, T. Luhman, E. Luhman *et al.*, "Video generation models as world simulators," *OpenAI Blog*, vol. 1, no. 8, p. 1, 2024.

[6] T. Wan, A. Wang, B. Ai, B. Wen, C. Mao, C.-W. Xie, D. Chen, F. Yu, H. Zhao, J. Yang *et al.*, "Wan: Open and advanced large-scale video generative models," *arXiv preprint arXiv:2503.20314*, 2025.

[7] B. Kang, Y. Yue, R. Lu, Z. Lin, Y. Zhao, K. Wang, G. Huang, and J. Feng, "How far is video generation from world model: A physical law perspective," *arXiv preprint arXiv:2411.02385*, 2024.

[8] D. Ha and J. Schmidhuber, "World models," *arXiv preprint arXiv:1803.10122*, vol. 2, no. 3, 2018.

[9] D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi, "Dream to control: Learning behaviors by latent imagination," *arXiv preprint arXiv:1912.01603*, 2019.

[10] M. Yang, Y. Du, K. Ghasemipour, J. Tompson, D. Schuurmans, and P. Abbeel, "Learning interactive real-world simulators," *arXiv preprint arXiv:2310.06114*, 2023.

[11] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.

[12] S. Kou, L. Gan, D. Wang, C. Li, and Z. Deng, "Bayesdiff: Estimating pixel-wise uncertainty in diffusion via bayesian inference," *arXiv preprint arXiv:2310.11142*, 2023.

[13] M. Jazbec and W.-T. et al., "Generative uncertainty in diffusion models," in *Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence*, ser. Proceedings of Machine Learning Research, S. Chiappa and S. Magliacane, Eds., vol. 286. PMLR, 21–25 Jul 2025, pp. 1837–1858.

[14] A. Bakhtin, L. van der Maaten, J. Johnson, L. Gustafson, and R. Girshick, "Phyre: A new benchmark for physical reasoning," *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[15] L. Kaufman and P. J. Rousseeuw, *Finding groups in data: an introduction to cluster analysis*. John Wiley & Sons, 2009.

[16] ModelScope Community, "Diffsynth-studio," GitHub repository, 2026, accessed: 2026-01-29. [Online]. Available: https://github.com/modelscope/DiffSynth-Studio